Please read the instructions below before the exam starts, and listen carefully to any instructions given by the invigilators. You may read the question paper before the exam starts but you must not write anything until the invigilator says that you may start.

You will be given five minutes after the exam has finished to complete the front of any answer books used.

May/June 2023 **EC115 2022/3 A 800**

Formula Sheet with Statistical Table
Two Answer books

Only Casio FX-83GTPLUS or FX-85TGPLUS or FX-83GTx or FX-85GTx Casio FX- 85MS calculators are permitted

UNIVERSITY OF READING

Introductory Quantitative Methods in Economics and Business (EC115)

Two hours

The paper is divided into **TWO SECTIONS.**

Answer **ALL** questions in Section A (40 marks – 10 marks each question); and answer **TWO** questions in Section B (60 marks – 30 marks each question)

Section A

1. Expected value

A video arcade is running a new game that has the following prizes and probabilities. Table 1 below shows wins and losses for the videogame players given the new game and the respective probabilities.

Table 1: Probability Distribution of Prizes

X	P(X=x)
-25	0.2
-15	0.5
35	0.1
12	0.6

(a) Calculate the expected value of this game.

(3 marks)

(b) Would it be profitable for the video arcade in the long run to offer this game? Why?

(2 marks)

(c) Calculate the variance and the standard deviation of this game.

(3 marks)

(d) A new manager decides to increase all outcomes (both positive and negative) by 40%. Calculate the new expected value and variance of the above game. Was this a convenient move for the video arcade in the long run? Why?

(2 marks)

2. Aggregate data

A company that specialises in hospital equipment rewards agents who achieve their sales target with an annual bonus.

Table 2 below provides summary data about the distribution of bonuses and agents by type of product sold.

Table 2: Average bonus by product sold

Products sold	Number of agents	Average annual bonus (in £)
Electrocardiogram machines	25	3000
Sterilizers	40	3123
Surgical stretchers	3	1950
Portable oxygen tanks	9	2000
Medical lasers	13	1890
Medical laboratory equipment	2	4000

(a) What is the average bonus paid by the company?

(3 marks)

(b) Compute the variance and the standard deviation of bonuses and indicate its unit of measure.

(5 marks)

(c) A competitor shows a standard deviation of £595. Comment on the dispersion of the two distributions.

(2 marks)

3. Probabilities of the normal distribution

Suppose X is a normally distributed random variable with a mean of 30 and a variance of 4. Obtain the following probabilities.

(a) $P(30 < \mathrm{X} < 35)$ (3 marks)

(b) $P(\mathrm{X} > 21)$. (2 marks)

(c) $P(\mathrm{X} < 40)$. (2 marks)

(d) Find b such that P(X<b) = 0.7 (3 marks)

4. Probabilities (10 points)

(a) Explain, *in non-technical terms*, the distinctions between marginal, joint and conditional probabilities.

(3 marks)

(b) The Quantum organisation reports that in the 2016 US presidential election between Donald Trump and Hillary Clinton, Trump won 56% of the female vote and 46% of the male vote (www.quantum.com). These are conditional probabilities. For example, the probability that a person voted for Trump given that they are female is 0.56. It can be assumed that the voting population is divided equally between men and women, so that the probability of a person being female is 0.5 as is the probability of being male. Let T be the event of a vote for Trump and F be the event that a person is female.

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(i) Calculate all joint probabilities $\mathrm{P}(\mathrm{T},\mathrm{F})$, $\mathrm{P}(\overline{\mathrm{T}},\mathrm{F})$, $\mathrm{P}(\mathrm{T},\overline{\mathrm{F}})$, $\mathrm{P}(\overline{\mathrm{T}},\overline{\mathrm{F}})$ where a bar above a letter indicates the complementary event and calculate all marginal probabilities in a contingency table.

(4 marks)

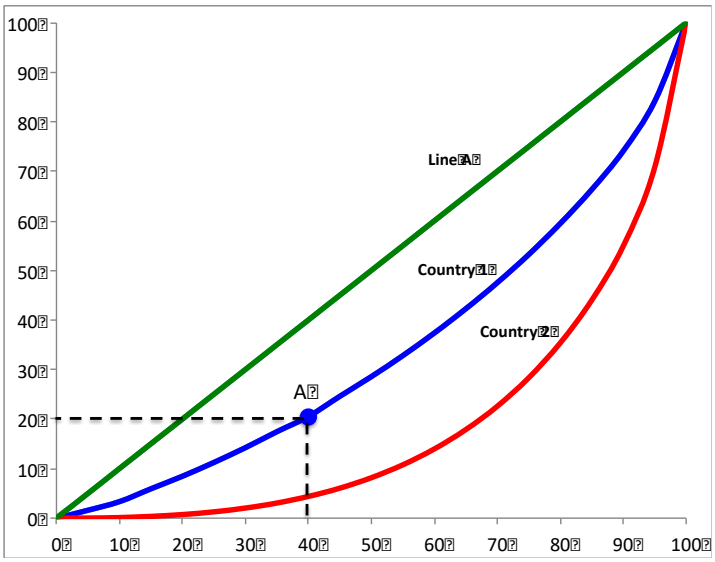
(ii) Calculate the probability that someone, regardless of gender, voted Trump. By what margin did Trump win the election?

(3 marks)

Section B

5. Inequality and the Gini coefficient

(a) Consider the following graph.



(i) Label and describe the axes

(ii) Name and describe the 3 different lines in the graph.
(iii) Explain what point A indicates.
(iv) Explain how the Gini coefficient of inequality can be computed using the above graph.
(v) How does inequality differ between the two countries?

You should provide comprehensive answers that show your understanding of the graph.

(15 marks)

(b) Consider Table 3 below that gives the proportion of total income by each quintile of the population from the poorest to the richest in a hypothetical country for the years 2000 and 2022.

Table 3: Income Distribution by Population Quintile

Quintile (from poorest to richest)	**Total income of households in each quintile**
1	40,000
2	46,000
3	68,000
4	120,000
5	190,000

(i) Calculate the Gini coefficient.

(10 marks)

(ii) How, if at all, would you expect the values of the Gini coefficient to change if the population was divided into deciles? Explain.

(5 marks)

6. Conditional probabilities

In March 2020 about 5 out of 1000 people living in the UK had contracted Covid19 (D). The most common diagnosis test at the time was correct at 80% for people with Covid19 (true positive)**.** Instead, the probability of false positive was 3%.

(a) Find the probability of the test being positive: P(T)

(6 marks)

(b) Find the probability of having contracted Covid19 for a random person who got a positive test in May 2020: P(D|T). Why does this probability differ substantially from P(T|D)?

(6 marks)

(c) In March 2021 the number of affected people in UK went up to 35 out of 1000 people? Find the probability of having contracted Covid19 for a random person who got a positive test in March 2021: P(D|T).

(6 marks)

(d) Consider a new test that gives more true positives (85%) but also more false positive (9%). Find the probability of having contracted Covid19 for a random person who got a positive outcome on this new test in March 2021. Provide an intuition to explain how P(D|T) calculated here differs from the one in c).

(6 marks)

(e) What if the disease was instead very uncommon with P(D) = 0.00158 (0.158%)? Recalculate P(D|T).

(6 marks)

7. Data distribution and random variables

(a) Table 4 below presents grouped data showing the index of real GDP for the following European countries at the end of 1998.

Table 4 – Index of real GDP

Index of real GDP	Frequency
55 to 64	**11**
65 to 74	**9**
75 to 84	**4**
85 to 94	**5**
95 to 104	**9**
105 to 114	**5**

(i) Calculate the group mean of the index of GDP.

(3 marks)

(ii) Estimate the median and mode from this frequency distribution using the class marks of the appropriate intervals. (3 marks)

(iii) Calculate the group variance from this distribution.
(3 marks)

(iv) Calculate the standard deviation.
(2 marks)

(v) Calculate the coefficient of variation.
(2 marks)

(vi) Calculate Bowleys coefficient of skewness for this data.
(2 marks)

(b) Table 5 below shows the probability distributions of revenues from two investment projects.

Table 5: Probability Distributions of Revenues (£ millions) from Two Investment Projects

Project A		Project B	
Revenue, r	**P(R=r)**	**Revenue, r**	**P(R=r)**
10	0.5	13	0.1
40	0.3	17	0.1
55	0.2	31	0.8

(i) Calculate the expected value of revenue for each project.
(4 marks)

(ii) Calculate the variance of revenue for each project.
(4 marks)

(iii) Which project, if either, is preferred? Explain.

(2 marks)

There are also costs associated with each project and revenues are taxed.
The profits of the project are the post-tax revenue minus the costs. The costs and tax rate of each project are different. Table 6 below gives the costs and revenue tax rates for each project.

Table 6: Costs (£ millions) and Tax Rates (%) of the Project

Project	Costs (£ millions)	Tax Rate (%)
A	7	20
B	5	10

(c) Let the letter G represent profits, t the tax rate (in %), and C costs. What is the equation for G in terms of C, t, and revenue, R?

(2 mark)

(d) Use your equation obtained in part (d) to calculate the expected value and variance of profits of the two projects.

(2 marks)

(e) Which project is preferred on grounds of profits? Explain.

(1 mark)

8. Inference

Consider this sample of employees in Company X, presented in Table 7.

Table 7: Hours spent working by employees of Company X:

Employee ID	Hours spent working
ID1	2
ID2	6
ID3	8
ID4	2
ID5	7
ID6	9
ID7	5
ID8	5
ID9	10
ID10	8
ID11	8
ID12	9
ID13	12
ID14	1
ID15	4
ID16	5
ID17	9
ID18	2
ID19	0
ID20	5
ID21	10
ID22	0
ID23	4
ID24	2

(a) Consider the possibility of using this sample to make some inference about the average number of hours that employees at Company X spent on work. What estimator would you use? How is your estimator distributed? State the assumptions you used to answer the question.

(5 marks)

(b) Do employees at Company X work as much as recommended by company guidelines, i.e. 6 hours per week? Test the hypothesis that these employees work on average as much as recommended by company guidelines using a 99% confidence interval. For this answer assume that the population is normally distributed, and that the variance is known and equal to 8. Show the workings and interpret your findings.

(5 marks)

(c) Instead of accepting any prior view on the variance, use the sample variance of 11.998 instead. Explain in words how the procedure you are going to use differs from the one used in b) and calculate the 99% confidence interval.

(6 marks)

(d) Obtain the same answer as above by writing down the null and alternative hypothesis and provide a statistical test to support the claim at 1% significance level. Show the workings and interpret your findings.

(10 marks)

(e) If the same average and variance used above in c) were obtained from a sample of 5 students, how would the confidence interval change? How would the standardised sample mean and its distribution change?

(4 marks)

(End of Examination Paper)